

What characteristics does Good Software have?

Vevox poll

<https://tudublin.vevox.com/#/present/274885>

Course overview

- What is the scope of Quality Assurance?
- Metrics and process improvement
- Software development processes
- Testing for correctness **(including lab on unit testing)**
- Change management and versioning **(including labs on Git and Gitflow)**
- Dealing with defects **(including lab on Peer review and Pull Requests)**

Assessment

- 30% continuous assessment
 - Group project using Scrum, Git, Github Flow, Pull requests
- 70% examination

Week #	Topic	Assessment
1	Introduction to SQA	
2	Processes, metrics, goals	
3	Configuration Management	
4	Git and Github Flow (lab)	
5	Software Defects	
6	Pull Requests and Code Reviews (lab)	Group CA begins
7	Testing	
8	Unit testing (lab)	Group CA due
9	Test coverage (lab)	
10	Software Development Processes	
11	Scope of SQA	
12		

What do we mean by Software
Quality?

What is software "Quality"?

- Zero defects
- The degree to which a system, component, or process meets customer or user needs or expectations
- Quality is just not the absence of defects (Backward looking) But the presence of value (Forward looking). Looking only at the quality of conformance gives us a mind set of ABSENCE OF PROBLEMS as the aim

Why is it challenging to build high quality software?

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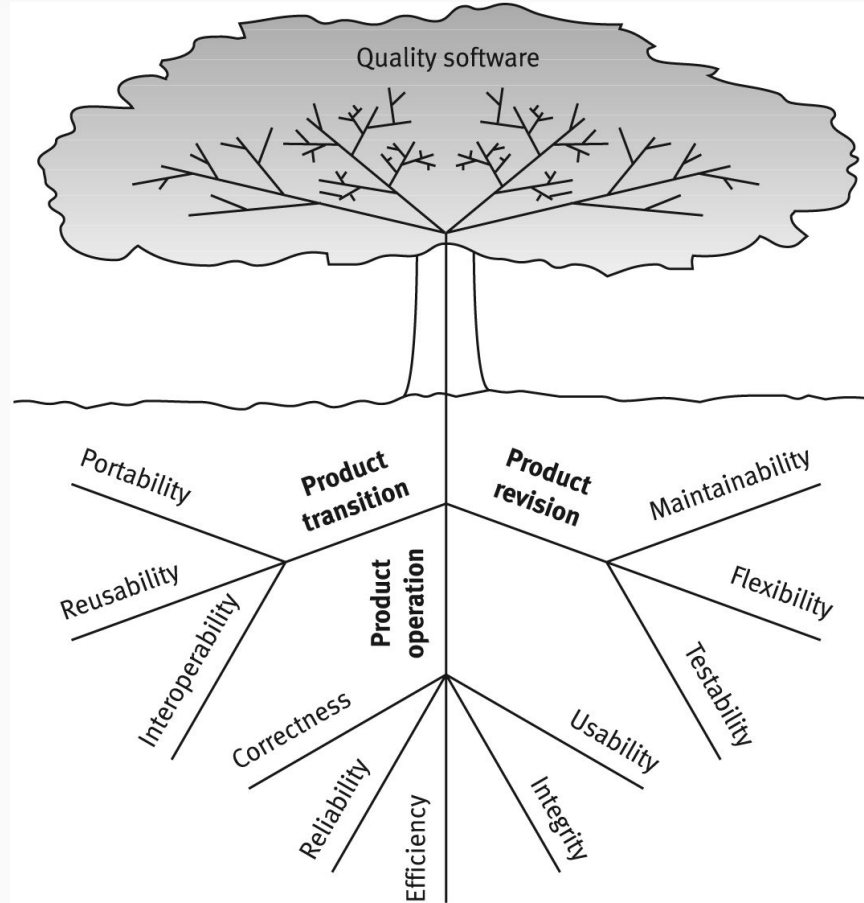
- High complexity of software systems
- Large scope
- Multiple contributors
- Limited opportunities to detect defects

Defining Software Quality

Various attempts to define Quality

- McCalls Factor Model Tree
 - 1977!
- International Standards Organisation (ISO) standard 25010
 - 2011

McCalls Factor Model Tree



ISO 25010 Systems and software Quality Requirements and Evaluation



<https://iso25000.com/index.php/en/iso-25000-standards/iso-25010>

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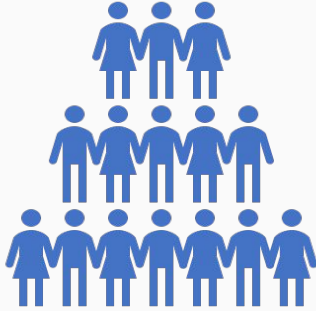
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Checking in



How is everyone doing?



Good / OK / Confused

Software Quality Assurance (SQA or QA)

What is Quality Assurance in Software?

- A planned and systematic pattern of all actions necessary to provide adequate confidence that an item or product conforms to established technical requirements
- Conformance to explicitly stated functional and performance requirements, explicitly documented development standards and implicit characteristics that are expected of all professionally developed software

Who's involved?

Stakeholders care about Quality

- **Product and project managers** - It will not be possible to perfectly satisfy the requirements of every stakeholder, and this group negotiates tradeoffs that are acceptable to principal stakeholders
- **Software engineers** - Responsible for building the product in line with the requirements

Stakeholders care about Quality

- **Users** - those who will operate the software, in various roles and with differing requirements
- **Customers** - those who have paid for the software
- **Regulators** - Many application domains, such as banking and public transport, are regulated. Software in these domains must comply with regulations e.g. on privacy or encryption

Barriers to Software Quality

Some problems occur in most/all projects

- Poor requirements
 - unclear, incomplete, too general, not testable, deviated from
- Unrealistic Schedule
 - too much work, too little time
- Inadequate Testing
- Poor communication between stakeholders

Less common barriers

- **Thinking that "Our problems are different"**

Software quality problems are not unique or uncommon

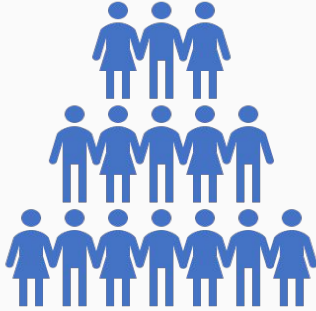
- **New hardware/software will transform software development**

Quality and productivity comes from people, not equipment/tools

- **"That's good enough - we don't have time to do better"**

Doing the right things the first time IS faster in the long term

Checking in



How is everyone doing?



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Review

Today

- Overview of the module
- What do we mean by Quality?
- Can we formally define it?
- What are the barriers to achieving high quality?